

Arpan Pathak

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SUMMARY

Software Engineer with nearly 8 years of industry experience architecting high-performance **database internals**, **cloud infrastructure**, **distributed systems**, and real-time **ML inference** pipelines at **Amazon**, **Microsoft**, and **Oracle**. Deep expertise in **Kotlin (JVM)**, **Rust**, **C++**, **Go**, and **Python**: lock-free concurrency, memory safety, event-driven distributed systems using **Kafka**, big data processing in **Apache Spark** and **AWS Glue**, and ML inference at scale. Shipped high-throughput, resource-efficient systems at global scale, from lock-free database engines and cloud control planes to real-time ML ranking and petabyte-scale data platforms.

PERSONAL PROJECTS

Driving-CivicSense (Edge AI Vision) | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- On-device AI perception system built in **Rust** for intersection discipline and lane awareness, running entirely on-device (< **50ms inference latency**, no cloud dependency) for aftermarket dashcam and AR-glasses form factors.
- Fine-tunes **YOLOv8/v11** and custom **CNN** architectures; applies model pruning and quantization to fit tight memory and power budgets on low-power ARM single-board computers (Raspberry Pi Zero, Raspberry Pi 5, and other low-power ARM processor boards).
- Trains on cloud GPU infrastructure with synthetic data augmentation; planning to release a public driving-behavior dataset on **Hugging Face**.

Companion Book: Seeing Machines: Deep Learning & Computer Vision from Python to Bare Metal | arpanpathak.github.io/seeing-machines-book

Physics & Mathematics of Game Development in Rust (Book) | arpanpathak.github.io/bevy-physics-book | 2026

EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

Jun 2025 - Aug 2026

- Rust Migration (Legacy C++/C#)**: Spearheaded migration of the Information Protection service from a legacy **unsafe C++** and garbage-collected **C#** codebase to **Rust**, shifting memory safety to compile-time. Completely eliminated **use-after-free**, **buffer-overflow**, and **null-safety** bugs in the migrated path. Delivered measurable performance gains over C#, **~40% lower P99 latency**, **~2x throughput**, and **~3x lower memory consumption per request**, with **6-figure annual cost savings**. Significantly increased developer productivity and confidence writing memory-safe code for MSEC.
- AI-Powered Policy Engine**: Built an AI-driven engine that auto-generates Cilium network policies and scans for misconfigurations, reducing manual policy review cycles from **4 hours** to near-real-time with zero policy drift across sovereign regions.
- Zero-Trust Sovereign Cloud Networking**: Architected **zero-trust sovereign cloud networking** using **eBPF** and **Cilium-based network policies** to ensure **data residency** and **SecNumCloud** compliance standards. Built **Kubernetes operators** to auto-generate network policies with CI/CD pipelines.
- CI/CD & Observability**: Established CI/CD pipelines, containerized microservices, and automated validation frameworks ensuring reliable multi-region releases. Integrated observability with **Prometheus** and **Azure Monitor**, and implemented **blue-green deployments**, improving operational efficiency.

TechStack: Rust, C++, Kubernetes, eBPF

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 - Mar 2025

- High-Performance Engine**: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to reduce contention in high-concurrency read-write paths, improving transaction throughput by **45,000 ops/s**.
- Infrastructure-as-Code SDK**: Engineered a Terraform provider (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control plane infrastructure with projected **\$1.2M/month cost savings**.
- Security & Key Management**: Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.

TechStack: Java, Go, Terraform Provider, Oracle Autonomous Database

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 - Oct 2024

- Real-Time ML Inference Engine**: Architected an **ensemble model pipeline combining XGBoost and Deep Learning** for real-time ads & deal ranking across FreeVee, Twitch, MiniTV, Prime Video, and Amazon Retail, processing millions of events at

100ms inference latency while driving **5% CTR increase**, **30%** higher deal impressions, and **\$5.4M monthly revenue growth**.

- **Event-Driven Ads Systems:** Designed and delivered event-driven systems for the ads publisher platform, delivering **programmatic guaranteed ad deals** via real-time **change data capture (CDC)** with **< 50ms latency** for ad consumer-side platforms handling auction and programmatic buying inventory.
- **Distributed Systems & SDK Design:** Designed a uniqueness constraint indexing solution and **SDK** for a multi-tenant NoSQL datastore using **two-phase commit**, achieving **serializable isolation** at **3K writes/second**, a developer-facing library used across multiple publisher teams.
- **Conversational AI (BERT):** Developed a **BERT-based inference service** handling **5,000 QPS** at peak, reducing support ticket triage SLA from **7 days** to **2 hours**.
- **Big Data Platform & Data Lake:** Built big data platform, data warehouse, and data lake infrastructure for Amazon WorkEvents data handling **1 petabyte** of data and serving OLTP use cases. Designed a **domain-specific language (DSL) execution engine** to deploy datasets to **AWS QuickSight** for business intelligence, using **AWS Glue**, **Kotlin Spark API**, and **AWS CDK (TypeScript)** to orchestrate job infrastructure and data refresh strategies.

TechStack: **Kotlin, Java, Python, TypeScript, AWS CDK, React.js, DynamoDB, Redshift, AWS, PostgreSQL, PyTorch, XGBoost**

Software Development Engineer, Razorpay, Bengaluru, India

Aug 2020 - Feb 2021

- Built UPI Payment Gateway Aggregator using **Golang** and Protobuf, processing **1M+ hourly transactions**.
- Maintained **99.99% uptime** with **150ms P99 latency** under **50,000 TPM** peak loads.

TechStack: **Go, Redis, PostgreSQL, Protobuf, Google Cloud, AWS**

Software Engineer, Mindfire Solutions, Bhubaneswar, India

Aug 2018 - Feb 2020

- Developed full-stack workflow tools and an **AR-based product search microservice**, serving 10,000+ daily queries.
- Built administrative dashboards reducing data retrieval latency from 5 min to 1 min.

TechStack: **Java, SpringBoot, Angular, MySQL, MongoDB, AWS**

SKILLS

Languages: Rust, C++, Go, Python, Java, Kotlin, C, Shell Script

Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems

Networking & Protocols: TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, WebSockets, Socket Programming, Network Performance Optimization

Network & Cloud Security: eBPF, Cilium, XDP, Kernel-Level Networking, Packet Processing, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

Distributed Systems: Distributed Systems (consensus, replication, sharding), Real-Time Data Pipelines, Microservices

Real-Time Streaming: Apache Kafka, Apache Flink

Big Data: Apache Spark, AWS Glue, AWS EMR

APIs & Serialization: gRPC, REST API Design, Protobuf

Frontend: React.JS, TypeScript, Angular

ML Inference & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Sequential Models, Natural Language Processing (NLP), Deep Learning, Recommendation Systems, MLOps

GPU & Parallel Computing: CUDA, CUDA-Oxide (actively exploring), GPU Kernel Programming, Parallel Computing, GPU-Accelerated Computing

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, SageMaker), Google Cloud, Azure (AKS), Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio

Observability: Prometheus, Grafana, Azure Monitor, AWS CloudWatch

Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Linux, Computer Architecture

EDUCATION

Bachelor of Technology in Computer Science & Engineering

RCC Institute of Information Technology, Kolkata, India | 2018